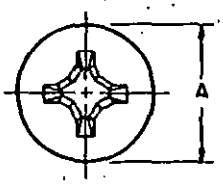
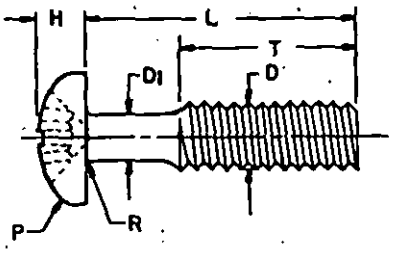
FED. SUP CLASS 5305





Ⓐ TABLE I. DASH NUMBERS AND DIMENSIONS.

D	NOMINAL SIZE		#4 .1120	#6 .1380	#8 .1640	#10 .1900	1/4 .2500	5/16 .3125
	THREADS PER INCH		40UNC-2A	32UNC-2A	32UNC-2A	24UNC-2A	20UNC-2A	18UNC-2A
D_1	RELIEF DIAMETER	MAX	.088	.106	.124	.143	.185	.229
		MIN	.078	.096	.114	.133	.175	.219
A	HEAD DIAMETER	MAX	.219	.270	.322	.373	.492	.615
		MIN	.205	.256	.306	.357	.473	.594
H	HEAD HEIGHT	MAX	.080	.097	.115	.133	.175	.218
		MIN	.070	.087	.105	.122	.162	.203
P	HEAD RADIUS	MIN	.010	.015	.015	.020	.035	.040
R	FILLET RADIUS	MAX	.017	.021	.025	.029	.038	.047
T	THREAD LENGTH	MAX	.196	.242	.287	.333	.438	.547
		MIN	.176	.222	.267	.312	.417	.527
L	LENGTH	TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
	3/8 (0.375)	+.000 -.031	1	--	--	--	--	--
	7/16 (0.438)		2	8	--	--	--	--
	1/2 (0.500)		3	9	--	--	--	--
	5/8 (0.625)		4	10	14	--	--	--
	3/4 (0.750)		5	11	15	19	24	--
	7/8 (0.875)		6	12	16	20	25	29
	1 (1.000)		7	13	17	21	26	30
	1-1/4 (1.250)	+.000 -.062	--	--	18	22	27	31
	1-1/2 (1.500)		--	--	--	23	28	32

REQUIREMENTS:

1. MATERIAL. STEEL, CORROSION-RESISTANT, 300 SERIES, AS SPECIFIED IN QQ-S-763 AND IN ACCORDANCE WITH FED-STD-66. STEEL, CORROSION-RESISTANT, 300 SERIES SHALL ALSO BE EQUAL TO OR INTERCHANGEABLE WITH 16:18 OR 18:8 CHROME-NICKEL ALLOY STEEL.
Ⓐ 2. FINISH. THE SCREWS SHALL BE BLACK OXIDE COATED IN ACCORDANCE WITH MIL-C-13924, CLASS 4, IF SPECIFIED, (SEE REQ. 7).
Ⓐ 3. CLEANING AND DESCALING. THE SCREWS SHALL BE CLEANED, DESCALED, AND PASSIVATED IN ACCORDANCE WITH ASTM A 380-78.
4. THREADS. THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/2. ACCEPTABILITY OF SCREW THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 21.
5. MAGNETIC PERMEABILITY. THE SCREWS SHALL HAVE A MAGNETIC PERMEABILITY OF 2.0 MAXIMUM (AIR=1.0) FOR A FIELD STRENGTH OF H=200 OERSTEDS USING A MAGNETIC INDICATOR IN ACCORDANCE WITH MIL-I-17214.
6. RECESS. THE SCREW RECESS SHALL BE IN ACCORDANCE WITH MS90006.
Ⓐ 7. PART NUMBER. THE PART NUMBER CONSISTS OF THE BASIC MS SHEET NUMBER FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE I. THE LETTER "B" FOR BLACK OXIDE COATING SHALL BE ADDED, IF SPECIFIED.

EXAMPLE: MS51896-1
- -1 — DASH NUMBER
- MS51896 — BASIC MS SHEET NUMBER

MS51896-1 = SCREW, EXTERNALLY RELIEVED BODY, PAN HEAD, CROSS-RECESSED, CRES, NOM. SIZE .1120-40UNC-2A, LENGTH 0.375.

NOTES:

1. DIMENSIONS. ALL DIMENSIONS ARE IN INCHES.
2. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON THE DATE OF INVITATIONS FOR BIDS, OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.
3. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

Ⓐ DENOTES CHANGES

P. A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
AR		SCREW, EXTERNALLY RELIEVED BODY, PANHEAD, CROSS-RECESSED, CORROSION-RESISTING STEEL, UNC-2A	MS51896
Other Cust AS 99			
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES		PAGE 1 OF 1

DD FORM 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5305-1507

USER ACTIVITIES: ARMY-AT, ME, MI NAVY-OS

REVIEWER ACTIVITIES: ARMY-AV, EA AIR FORCE-82 DLA-IS

APPROVED 29 AUG 1977 REVISED Ⓐ 21 JUL 83

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:12Z
Check the source to verify that this is the current version before use.